Bisimulation is the resized identity of the cumulative hierarchy

The Univalent Foundations Program

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Editorial note. Original author statement, proof and supporting context preserved. Only the document wrapper and navigational reference bindings are added; no mathematical repair or correctness certification.

Allowed standard prerequisites (editorial).

- Homotopy type theory. Univalence, dependent products and sums, identity types, propositional and n-truncation; native book references retained.. Standard background, without replacing the counted argument.

Source reference. Original reference 10.5.7: combinatorics-logic/sources/hott-book/main.labelnumbers.first-edition, source lines 1192–1192. Reference to book navigation or standard prerequisite; the target author proof is retained in full.

Source reference. Original reference 10.11: combinatorics-logic/sources/hott-book/main.labelnumbers.first-edition, source lines 1351–1351. Reference to book navigation or standard prerequisite; the target author proof is retained in full.

Source reference. Original reference 6.9: combinatorics-logic/sources/hott-book/main.labelnumbers.first-edition, source lines 1663–1663. Reference to book navigation or standard prerequisite; the target author proof is retained in full.

Source reference. Original reference 6.13: combinatorics-logic/sources/hott-book/main.labelnumbers.first-edition, source lines 1685–1685. Reference to book navigation or standard prerequisite; the target author proof is retained in full.

10.1 The cumulative hierarchy

We can define a cumulative hierarchy V of all sets in a given universe $\mathcal{U}$ as a higher inductive type, in such a way that V is again a set (in a larger universe $\mathcal{U}'$), equipped with a binary "membership" relation $x \in y$ which satisfies the usual laws of set theory.

Definition 10.1.1. The **cumulative hierarchy** V relative to a type universe $\mathcal{U}$ is the higher inductive type generated by the following constructors.

(i) For every $A : \mathcal{U}$ and $f : A \to V$, there is an element $\mathsf{set}(A, f) : V$.

(ii) For all $A, B : \mathcal{U}$, $f : A \to V$ and $g : B \to V$ such that

$$\big(\forall(a : A).\, \exists(b : B).\, f(a) =_V g(b)\big) \wedge \big(\forall(b : B).\, \exists(a : A).\, f(a) =_V g(b)\big) \tag{10.1.2}$$

there is a path $\mathsf{set}(A, f) =_V \mathsf{set}(B, g)$.

(iii) The 0-truncation constructor: for all $x, y : V$ and $p, q : x = y$, we have $p = q$.

In set-theoretic language, $\mathsf{set}(A, f)$ can be understood as the set (in the sense of classical set theory) that is the image of A under f, i.e. $\{\, f(a) \mid a \in A \,\}$. However, we will avoid this notation, since it would clash with our notation for subtypes (but see (10.1.3) and original reference 10.5.7 below).

The hierarchy V is bootstrapped from the empty map $\mathsf{rec}_{\mathbf{0}}(V) : \mathbf{0} \to V$, which gives the empty set as $\emptyset = \mathsf{set}(\mathbf{0}, \mathsf{rec}_{\mathbf{0}}(V))$. Then the singleton $\{\emptyset\}$ enters V through $\mathbf{1} \to V$, defined as $\star \mapsto \emptyset$, and so on. (The definition can also be adapted to include an arbitrary set of "atoms" or "urelements", by adding an additional point constructor.) The type V lives in the same universe as the base universe $\mathcal{U}$.

The second constructor of V has a form unlike any we have seen before: it involves not only paths in V (which in original reference 6.9 we claimed were slightly fishy) but truncations of sums of them. It certainly does not fit the general scheme described in original reference 6.13, and thus it may not be obvious what its induction principle should be. Fortunately, like our first definition of the 0-truncation in original reference 6.9, it can be re-expressed using auxiliary higher inductive types. We leave it to the reader to work out the details (see original reference 10.11).

At the end of the day, the induction principle for V (written in pattern matching language) says that given $P : V \to \mathsf{Set}$, in order to construct $h : \prod_{(x:V)} P(x)$, it suffices to give the following.

(i) For any $f : A \to V$, construct $h(\mathsf{set}(A, f))$, assuming as given $h(f(a))$ for all $a : A$.

(ii) Verify that if $f : A \to V$ and $g : B \to V$ satisfy (10.1.2), then $h(\mathsf{set}(A, f)) =^P_q h(\mathsf{set}(B, g))$, where q is the path arising from the second constructor of V and (10.1.2), assuming inductively that $h(f(a))$ and $h(g(b))$ are defined for all $a : A$ and $b : B$, and that the following condition holds:

$$\begin{aligned} &\big(\forall(a : A).\, \exists(b : B).\, \exists(p : f(a) = g(b)).\, h(f(a)) =^P_p h(g(b))\big) \\ \wedge\quad &\big(\forall(b : B).\, \exists(a : A).\, \exists(p : f(a) = g(b)).\, h(f(a)) =^P_p h(g(b))\big) \end{aligned}$$

The second clause checks that the map being defined must respect the paths introduced in (10.1.2). As usual when we state higher induction principles using pattern matching, it may seem tautologous, but is not. The point is that "$h(f(a))$" is essentially a formal symbol which we cannot peek inside of, which $h(\mathsf{set}(A, f))$ must be defined in terms of. Thus, in the second clause, we assume equality of these formal symbols when appropriate, and verify that the elements resulting from the construction of the first clause are also equal. Of course, if P is a family of mere propositions, then the second clause is automatic.

Observe that, by induction, for each $v : V$ there merely exist $A : \mathcal{U}$ and $f : A \to V$ such that $v = \mathsf{set}(A, f)$. Thus, it is reasonable to try to define the

membership relation $x \in v$ on V by setting:

$$(x \in \mathsf{set}(A, f)) :\equiv (\exists(a : A).\, x = f(a)).$$

To see that the definition is valid, we must use the recursion principle of V. Thus, suppose we have a path $\mathsf{set}(A, f) = \mathsf{set}(B, g)$ constructed through (10.1.2). If $x \in \mathsf{set}(A, f)$ then there merely is $a : A$ such that $x = f(a)$, but by (10.1.2) there merely is $b : B$ such that $f(a) = g(b)$, hence $x = g(b)$ and $x \in \mathsf{set}(B, g)$. The converse is symmetric.

The **subset relation** $x \subseteq y$ is defined on V as usual by

$$(x \subseteq y) :\equiv \forall(z : V).\, z \in x \Rightarrow z \in y.$$

A **class** may be taken to be a mere predicate on V. We can say that a class $C : V \to \mathsf{Prop}$ is a V**-set** if there merely exists $v : V$ such that

$$\forall(x : V).\, C(x) \Leftrightarrow x \in v.$$

We may also use the conventional notation for classes, which matches our standard notation for subtypes:

$$\{\, x \mid C(x) \,\} :\equiv \lambda x.\, C(x). \tag{10.1.3}$$

A class $C : V \to \mathsf{Prop}$ will be called $\mathcal{U}$**-small** if all of its values $C(x)$ lie in $\mathcal{U}$, specifically $C : V \to \mathsf{Prop}_{\mathcal{U}}$. Since V lives in the same universe $\mathcal{U}'$ as does the base universe $\mathcal{U}$ from which it is built, the same is true for the identity types $v =_V w$ for any $v, w : V$. To obtain a well-behaved theory in the absence of propositional resizing, therefore, it will be convenient to have a $\mathcal{U}$-small "resizing" of the identity relation, which we can define by induction as follows.

Definition 10.1.4. Define the **bisimulation** relation

$$\sim : V \times V \longrightarrow \mathsf{Prop}_{\mathcal{U}}$$

by double induction over V, where for $\mathsf{set}(A, f)$ and $\mathsf{set}(B, g)$ we let:

$$\mathsf{set}(A, f) \sim \mathsf{set}(B, g) :\equiv \big(\forall(a : A).\, \exists(b : B).\, f(a) \sim g(b)\big) \wedge \big(\forall(b : B).\, \exists(a : A).\, f(a) \sim g(b)\big).$$

To verify that the definition is correct, we just need to check that it respects paths $\mathsf{set}(A, f) = \mathsf{set}(B, g)$ constructed through (10.1.2), but this is obvious, and that $\mathsf{Prop}_{\mathcal{U}}$ is a set, which it is. Note that $u \sim v$ is in $\mathsf{Prop}_{\mathcal{U}}$ by construction.

Lemma 10.1.5. *For any $u, v : V$ we have $(u =_V v) = (u \sim v)$.*

Proof. An easy induction shows that $\sim$ is reflexive, so by transport we have $(u =_V v) \to (u \sim v)$. Thus, it remains to show that $(u \sim v) \to (u =_V v)$. By induction on u and v, we may assume they are $\mathsf{set}(A, f)$ and $\mathsf{set}(B, g)$ respectively. (We can ignore the path constructors of V, since $(u \sim v) \to (u =_V v)$ is a mere proposition.) Then by definition, $\mathsf{set}(A, f) \sim \mathsf{set}(B, g)$ implies $(\forall(a : A).\, \exists(b : B).\, f(a) \sim g(b))$ and conversely. But the inductive hypothesis then tells us that $(\forall(a : A).\, \exists(b : B).\, f(a) = g(b))$ and conversely. So by the path constructor for V we have $\mathsf{set}(A, f) =_V \mathsf{set}(B, g)$. □